

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTBSC404 - PROBABILITY AND STATISTICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 A variable whose value is determined by the outcome of a random experiment is called
i) Random Variable ii) Attribute iii) constant iv) none of these
- 2 The correlation coefficient between two variables X and Y is found to be 0.4. What is the correlation between 2X and (-Y)?
i) -0.5 ii) 0.4 iii) 0.5 iv) -0.4
- 3 In a Poisson distribution, the variance is always:
i) greater than the mean ii) less than the mean
iii) equal to the mean iv) equal to the square root of the mean
- 4 How many four-letter computer passwords can be formed using only the symmetric letters (no repetition is allowed)?
i) 7920 ii) 390 iii) 1640 iv) 24
- 5 If X is Poisson variable such that $P(X=2) = 9P(X=4) + 90P(X=6)$, find the mean of X.
i) 2 ii) 1 iii) 4 iv) 3
- 6 The coefficient of correlation between debenture prices and share prices is found to be 0.143. If the sum of squares of differences in the rank is given to be 48. Find the value of n.
i) 8 ii) 3 iii) 7 iv) 9
- 7 A bag contains 8 black, 3 red, and 9 white balls. If 3 balls are drawn at random, find the probability that none is red.
i) 17/57 ii) 22/57 iii) 34/57 iv) 21/95
- 8 If the null hypothesis is false, then which of the following is accepted?
i) Null Hypothesis ii) Positive Hypothesis
iii) Negative Hypothesis iv) Alternative Hypothesis
- 9 A discrete random variable X follows a Poisson distribution with a mean of 2. What is the probability of $X = 0$?
i) e^{-2} ii) $2e^{-2}$ iii) 1/2 iv) 1

- 10 In a standard normal distribution, what are the mean and standard deviation?
 i) Mean = 1, Standard deviation = 0 ii) Mean = 0, Standard deviation = 1
 iii) Mean = 1, Standard deviation = 1 iv) Mean = 0, Standard deviation = 0
- 11 The rejection probability of the Null Hypothesis when it is true is called?
 i) Level of Confidence ii) Level of Significance
 iii) Level of Margin iv) Level of Rejection
- 12 What does the standard deviation of a normal distribution represent?
 i) The average value of the data.
 ii) The spread or dispersion of the data around the mean.
 iii) The peak of the distribution. iv) The total number of data points.

Q2. Solve the following.

- A) A quality control manager found that 30% of worker-related problems occurred on 6 Mondays, and that 20% occurred in the last hour of a day's shift. It was also found that 4% of worker related problems occurred in the last hour of Monday's shift.
 i) What is the probability that a worker-related problem that occurs on a Monday does not occur in the last hour of the day's shift?
 ii) Are the events 'Problem occurs on Monday' and 'Problem occurs in the last hour of the day's shift' statistically independent?
- B) In the bolt factory, machines A, B, and C manufacture, respectively, 20%, 30% and 6 50% of the total. Of their output, 5, 4, and 2 percent are known to be defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it was manufactured by
 (i) machine B (ii) machine A or C

Q3. Solve the following.

- A) i) For any Three random variables X_1, X_2, X_3 show that: 6

$$\text{cov}(X_1 + X_2, X_3) = \text{cov}(X_1, X_3) + \text{cov}(X_2, X_3).$$

 ii) Find the variance of the number obtained on a throw of an unbiased die.
- B) Two cards are drawn with and without replacement from a well shuffled pack of 6 cards. Obtain the probability distribution of the number of face cards (Jack, Queen, King, and Ace) with replacement and without replacement.

Q4. Solve Any Two of the following.

- A) Obtain an expression for MEAN and VARIANCE for the Binomial Distribution in terms of the number of trials and the probability of success. Hence, show that for the Binomial Distribution, the variance is less than the mean.
- B) Time taken by the crew of a company to construct a small bridge is a normal variate with a mean of 400 labour hours and a standard deviation of 100 labour hours.
 (i) What is the probability that the bridge gets constructed between 350 to 450 labour hours?
 (ii) If the company promises to construct the bridge in 450 labour hours or less and agrees to pay a penalty of Rs. 100 for each labour hour spent in excess of 450, what is the probability that a company pays a penalty of at least Rs. 2000?
- C) A manufacturer finds that the average demand per day for mechanics to repair his 6 new product is 1.5 over a period of a year, and demand per day is distributed as a Poisson variate. He employs two mechanics. On how many days in one year:
 (i) both mechanics would be free, (ii) some demand is refused.

Q5. Solve Any Two of the following.

- A) Define the terms: (i) Positive Correlation, (ii) Linear Correlation. 6
Calculate Spearman's rank coefficient of correlation between advertisement cost and sales from the data given below:

Advertisement	39	65	62	90	82	75	25	98	36	78
Sales	47	53	58	86	62	68	60	91	51	84

- B) The data about the sales and advertisement expenditure of a firm is given below: 6

	Sales (in Crores of Rs.)	Advertisement Expenditure (in Crores of Rs.)
Mean	40	6
Standard Deviation	10	1.5

Coefficient of correlation = $r = 0.9$.

- i) Estimate the likely sales for a proposed advertisement expenditure of Rs. 10 crores.
ii) What should be the advertisement expenditure if the firm proposes a sales target of 60 crores of rupees?

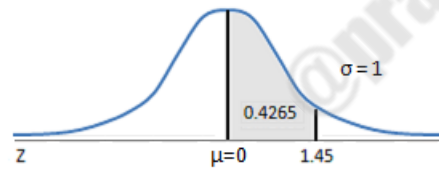
- C) Define the terms: (i) Negative Correlation, (ii) Correlation. 6
Prove that the limits of the Correlation Coefficient lie between $-1 \leq r \leq 1$.

Q6. Solve Any Two of the following.

- A) Distinguish between one tailed and two tailed tests, clearly bringing out the 6 differences between the alternative hypotheses under the two types of tests.
B) (a) In a sample of 400 parts manufactured by a factory, the number of defective parts was found to be 30. The company, however, claimed that almost 5% of its products are defective. Is the claim tenable?
(b) A sample of size 400 was drawn, and the sample mean was found to be 99. Test whether this sample could have come from a normal distribution with a mean of 100 and a variance of 64 at 5% level of significance.
C) i) The recipe for a bakery item is designed to result in a product that contains 8 grams of fat per serving. The quality control department samples the product periodically to ensure that the production process is working as designed. State the relevant null and alternative hypotheses.
ii) A publisher of college textbooks claims that the average price of all hardbound college textbooks is ₹127.50. A student group believes that the actual mean is higher and wishes to test their belief. State the relevant null and alternative hypotheses.

Areas Under the One-Tailed Standard Normal Curve

This table provides the area between the mean and some Z score.
For example, when Z score = 1.45 the area = 0.4265.



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

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